

Адресация IPv4 и IPv6. Настройка DHCP

Лабораторная работа №7

Чилеше Лупупа

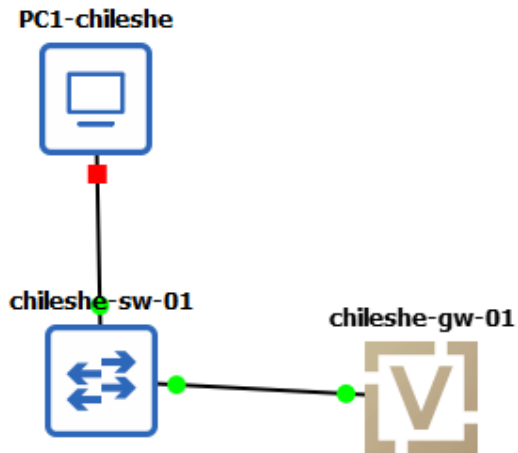
16 декабря 2025

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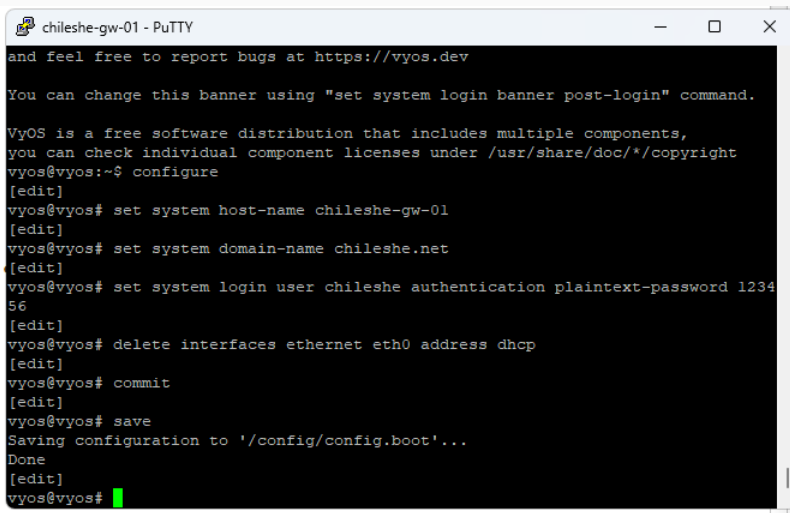
Цели и задачи работы

Получение навыков настройки службы DHCP на сетевом оборудовании для распределения адресов IPv4 и IPv6.

DHCP для IPv4 (VyOS + VPCS)



Первичная настройка VyOS



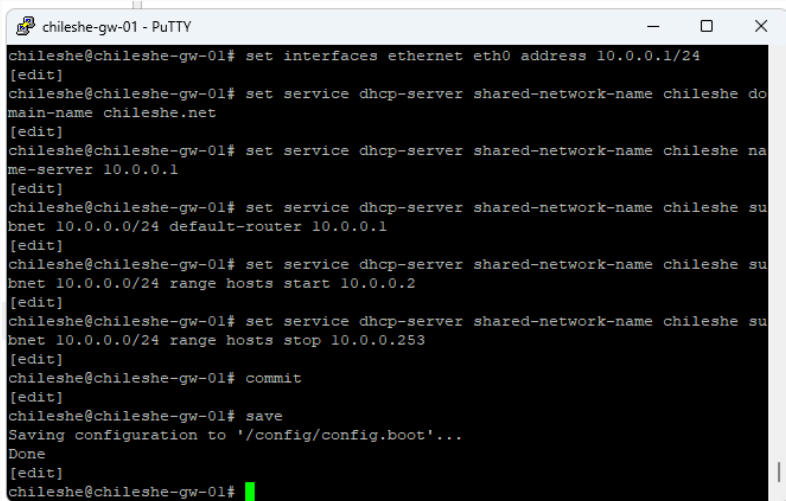
```
chileshe-gw-01 - PuTTY
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name chileshe-gw-01
[edit]
vyos@vyos# set system domain-name chileshe.net
[edit]
vyos@vyos# set system login user chileshe authentication plaintext-password 1234
56
[edit]
vyos@vyos# delete interfaces ethernet eth0 address dhcp
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos#
```

Рис. 2: Изменение host-name, domain-name, создание пользователя

Настройка IPv4 и DHCP-сервера



```
chileshe-gw-01 - PuTTY
chileshe@chileshe-gw-01# set interfaces ethernet eth0 address 10.0.0.1/24
[edit]
chileshe@chileshe-gw-01# set service dhcp-server shared-network-name chileshe do
main-name chileshe.net
[edit]
chileshe@chileshe-gw-01# set service dhcp-server shared-network-name chileshe na
me-server 10.0.0.1
[edit]
chileshe@chileshe-gw-01# set service dhcp-server shared-network-name chileshe su
bnet 10.0.0.0/24 default-router 10.0.0.1
[edit]
chileshe@chileshe-gw-01# set service dhcp-server shared-network-name chileshe su
bnet 10.0.0.0/24 range hosts start 10.0.0.2
[edit]
chileshe@chileshe-gw-01# set service dhcp-server shared-network-name chileshe su
bnet 10.0.0.0/24 range hosts stop 10.0.0.253
[edit]
chileshe@chileshe-gw-01# commit
[edit]
chileshe@chileshe-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
chileshe@chileshe-gw-01#
```

Рис. 3: IP 10.0.0.1/24 на eth0 и DHCP-пул 10.0.0.2–10.0.0.253

Получение IPv4 по DHCP на клиенте

```
PC1-chileshe - PuTTY
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Request
Option 54: DHCP Server = 10.0.0.1
Option 50: Requested IP Address = 10.0.0.2
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:50:79:66:68:00
Option 12: Host Name = PC1-chileshe

Opcode: 2 (REPLY)
Client IP Address: 10.0.0.2
Your IP Address: 10.0.0.2
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Ack
Option 54: DHCP Server = 10.0.0.1
Option 51: Lease Time = 86400
Option 1: Subnet Mask = 255.255.255.0
Option 3: Router = 10.0.0.1
Option 6: DNS Server = 10.0.0.1
Option 15: Domain = chileshe.net

IP 10.0.0.2/24 GW 10.0.0.1

PC1-chileshe> save
Saving startup configuration to startup.vpc
. done

PC1-chileshe> 
```



```
PC1-chileshe>
PC1-chileshe> show ip

NAME           : PC1-chileshe[1]
IP/MASK        : 10.0.0.2/24
GATEWAY        : 10.0.0.1
DNS            : 10.0.0.1
DHCP SERVER    : 10.0.0.1
DHCP LEASE     : 86377, 86400/43200/75600
DOMAIN NAME    : chileshe.net
MAC            : 00:50:79:66:68:00
LPORT          : 10010
RHOST:PORT     : 127.0.0.1:10011
MTU            : 1500

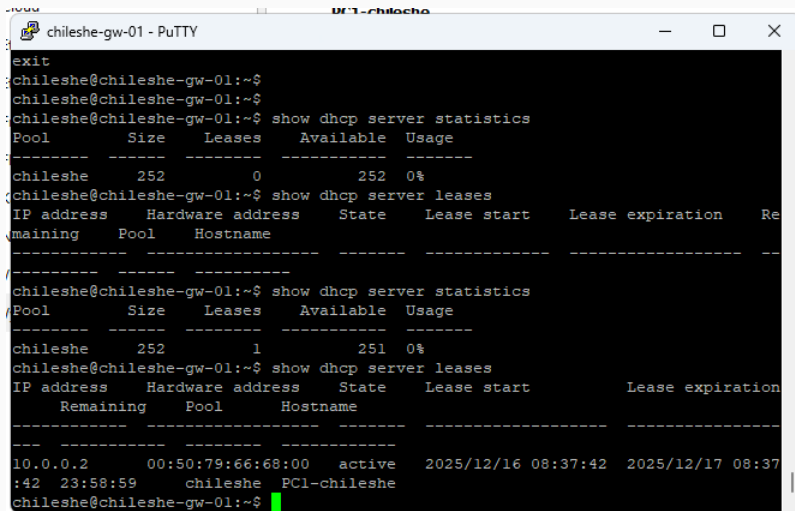
PC1-chileshe> ping 10.0.0.1 -c 2

84 bytes from 10.0.0.1 icmp_seq=1 ttl=64 time=3.684 ms
84 bytes from 10.0.0.1 icmp_seq=2 ttl=64 time=0.892 ms

PC1-chileshe> 
```

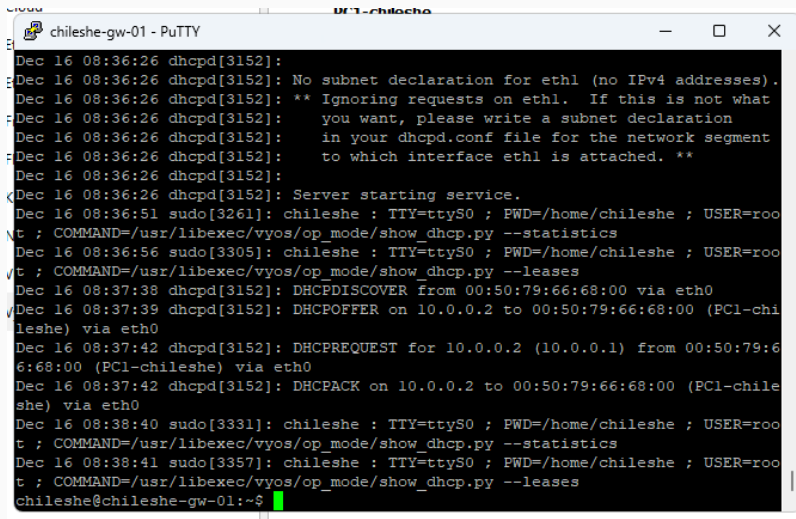
Рис. 5: show ip и ping 10.0.0.1

Статистика DHCP и выданные адреса



```
chileshe-gw-01 - PuTTY
exit
chileshe@chileshe-gw-01:~$
chileshe@chileshe-gw-01:~$
chileshe@chileshe-gw-01:~$ show dhcp server statistics
Pool          Size    Leases    Available  Usage
-----
chileshe      252      0         252        0%
chileshe@chileshe-gw-01:~$ show dhcp server leases
IP address    Hardware address  State    Lease start    Lease expiration  Re
maining      Pool    Hostname
-----
chileshe@chileshe-gw-01:~$ show dhcp server statistics
Pool          Size    Leases    Available  Usage
-----
chileshe      252      1         251        0%
chileshe@chileshe-gw-01:~$ show dhcp server leases
IP address    Hardware address  State    Lease start    Lease expiration
Remaining      Pool    Hostname
-----
10.0.0.2      00:50:79:66:68:00  active   2025/12/16 08:37:42  2025/12/17 08:37
:42 23:58:59    chileshe  PC1-chileshe
chileshe@chileshe-gw-01:~$
```

Рис. 6: show dhcp server statistics и show dhcp server leases



```
chileshe-gw-01 - PuTTY
Dec 16 08:36:26 dhcpd[3152]:
Dec 16 08:36:26 dhcpd[3152]: No subnet declaration for eth1 (no IPv4 addresses).
Dec 16 08:36:26 dhcpd[3152]: ** Ignoring requests on eth1.  If this is not what
Dec 16 08:36:26 dhcpd[3152]:     you want, please write a subnet declaration
Dec 16 08:36:26 dhcpd[3152]:     in your dhcpd.conf file for the network segment
Dec 16 08:36:26 dhcpd[3152]:     to which interface eth1 is attached. **
Dec 16 08:36:26 dhcpd[3152]:
Dec 16 08:36:26 dhcpd[3152]: Server starting service.
Dec 16 08:36:51 sudo[3261]: chileshe : TTY=ttyS0 ; PWD=/home/chileshe ; USER=root ; COMMAND=/usr/libexec/vyos/op_mode/show_dhcp.py --statistics
Dec 16 08:36:56 sudo[3305]: chileshe : TTY=ttyS0 ; PWD=/home/chileshe ; USER=root ; COMMAND=/usr/libexec/vyos/op_mode/show_dhcp.py --leases
Dec 16 08:37:38 dhcpd[3152]: DHCPDISCOVER from 00:50:79:66:68:00 via eth0
Dec 16 08:37:39 dhcpd[3152]: DHCPOFFER on 10.0.0.2 to 00:50:79:66:68:00 (PC1-chileshe) via eth0
Dec 16 08:37:42 dhcpd[3152]: DHCPREQUEST for 10.0.0.2 (10.0.0.1) from 00:50:79:66:68:00 (PC1-chileshe) via eth0
Dec 16 08:37:42 dhcpd[3152]: DHCPACK on 10.0.0.2 to 00:50:79:66:68:00 (PC1-chileshe) via eth0
Dec 16 08:38:40 sudo[3331]: chileshe : TTY=ttyS0 ; PWD=/home/chileshe ; USER=root ; COMMAND=/usr/libexec/vyos/op_mode/show_dhcp.py --statistics
Dec 16 08:38:41 sudo[3357]: chileshe : TTY=ttyS0 ; PWD=/home/chileshe ; USER=root ; COMMAND=/usr/libexec/vyos/op_mode/show_dhcp.py --leases
chileshe@chileshe-gw-01:~$
```

Рис. 7: События DHCPDISCOVER/DHCPOFFER/DHCPREQUEST/DHCPACK

Анализ DHCP-трафика

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	::	ff02::2	ICMPv6	62	Router Solicitation
2	18.281326	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0xd3c97809
3	18.293295	0c:56:73:6c:00:00	Broadcast	ARP	60	Who has 10.0.0.2? Tell 10.0.0.1
4	19.281473	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0xd3c97809
5	19.294621	10.0.0.1	10.0.0.2	DHCP	342	DHCP Offer - Transaction ID 0xd3c97809
6	19.297044	0c:56:73:6c:00:00	Broadcast	ARP	60	Who has 10.0.0.2? Tell 10.0.0.1
7	20.321251	0c:56:73:6c:00:00	Broadcast	ARP	60	Who has 10.0.0.2? Tell 10.0.0.1
8	22.281868	0.0.0.0	255.255.255.255	DHCP	406	DHCP Request - Transaction ID 0xd3c97809
9	22.288022	10.0.0.1	10.0.0.2	DHCP	342	DHCP ACK - Transaction ID 0xd3c97809
10	23.282250	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 10.0.0.2 (Request)

Frame 9: 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface -, id 0

Ethernet II, Src: 0c:56:73:6c:00:00 (0c:56:73:6c:00:00), Dst: Private_66:68:00 (00:50:79:66:68:00)

Internet Protocol Version 4, Src: 10.0.0.1, Dst: 10.0.0.2

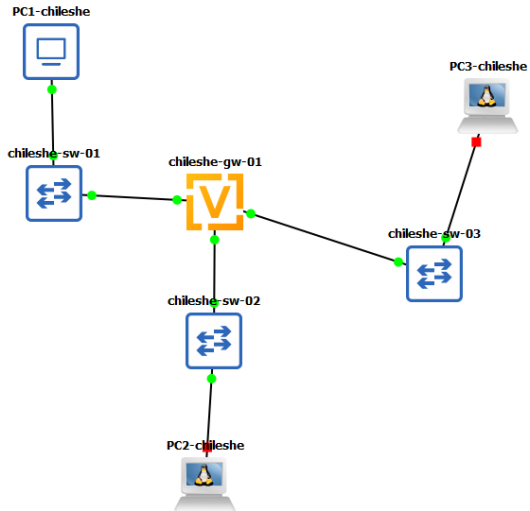
User Datagram Protocol, Src Port: 67, Dst Port: 68

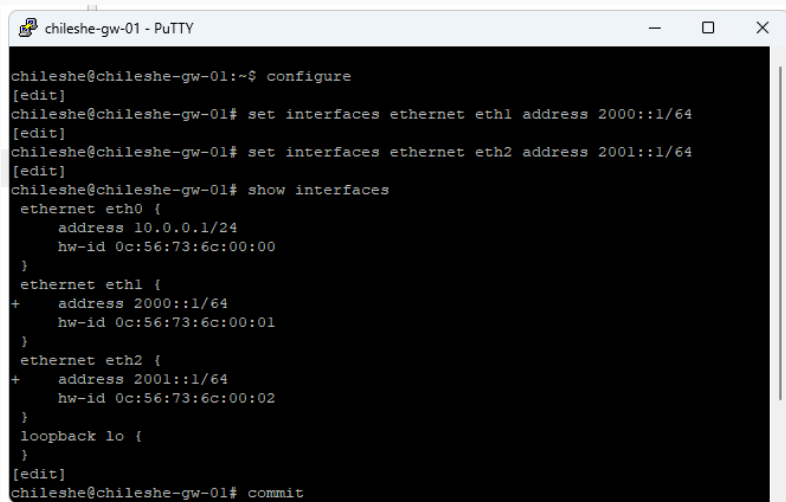
Source Port: 67
Destination Port: 68
Length: 308
Checksum: 0x7768 [unverified]
[Checksum Status: Unverified]
[Stream index: 1]
[Stream Packet Number: 2]
[Timestamps]
UDP payload (300 bytes)

Dynamic Host Configuration Protocol (ACK)

Message type: Boot Reply (2)
Hardware type: Ethernet (0x01)
Hardware address length: 6
Hops: 0
Transaction ID: 0xd3c97809
Seconds elapsed: 0
Bootp flags: 0x0000 (Unicast)
Client IP address: 10.0.0.2
Your (client) IP address: 10.0.0.2
Next server IP address: 0.0.0.0
Relay agent IP address: 0.0.0.0
Client MAC address: Private 66:68:00 (00:50:79:66:68:00)
Client hardware address padding: 00000000000000000000
Server host name not given
Boot file name not given
Magic cookie: DHCP
Option: (53) DHCP Message Type (ACK)
Option: (54) DHCP Server Identifier (10.0.0.1)
Option: (51) IP Address Lease Time
Option: (1) Subnet Mask (255.255.255.0)
Option: (3) Router
Option: (6) Domain Name Server
Option: (15) Domain Name
Option: (255) End
Padding: 000000000000000000000000

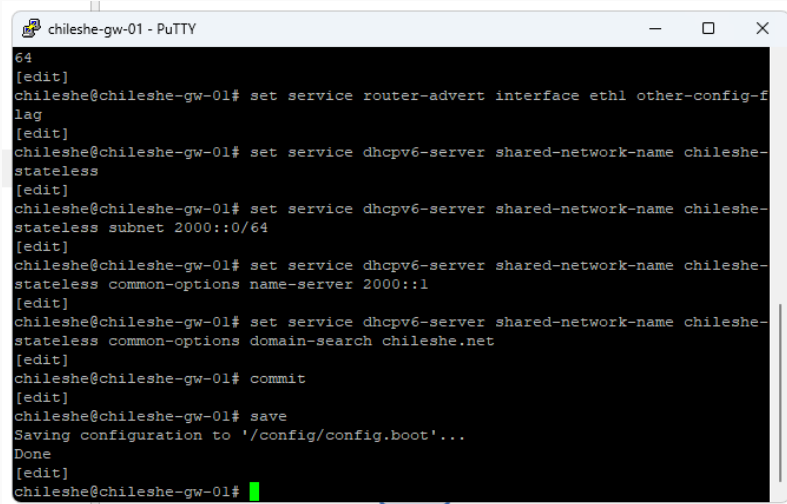
DHCPv6 Stateless





```
chileshe@chileshe-gw-01:~$ configure
[edit]
chileshe@chileshe-gw-01# set interfaces ethernet eth1 address 2000::1/64
[edit]
chileshe@chileshe-gw-01# set interfaces ethernet eth2 address 2001::1/64
[edit]
chileshe@chileshe-gw-01# show interfaces
  ethernet eth0 {
    address 10.0.0.1/24
    hw-id 0c:56:73:6c:00:00
  }
  ethernet eth1 {
+   address 2000::1/64
    hw-id 0c:56:73:6c:00:01
  }
  ethernet eth2 {
+   address 2001::1/64
    hw-id 0c:56:73:6c:00:02
  }
  loopback lo {
  }
[edit]
chileshe@chileshe-gw-01# commit
```

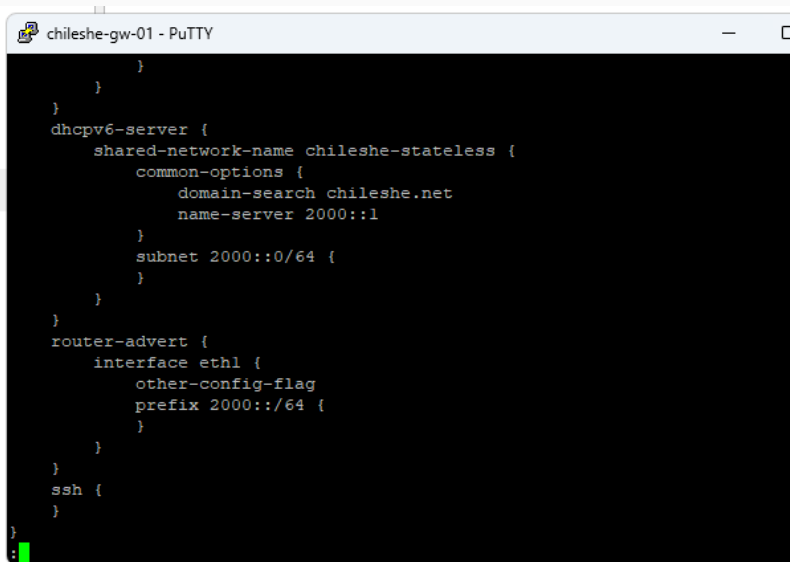
Рис. 10: eth1: 2000::1/64, eth2: 2001::1/64



```
chileshe-gw-01 - PuTTY
64
[edit]
chileshe@chileshe-gw-01# set service router-advert interface eth1 other-config-f
lag
[edit]
chileshe@chileshe-gw-01# set service dhcpv6-server shared-network-name chileshe-
stateless
[edit]
chileshe@chileshe-gw-01# set service dhcpv6-server shared-network-name chileshe-
stateless subnet 2000::0/64
[edit]
chileshe@chileshe-gw-01# set service dhcpv6-server shared-network-name chileshe-
stateless common-options name-server 2000::1
[edit]
chileshe@chileshe-gw-01# set service dhcpv6-server shared-network-name chileshe-
stateless common-options domain-search chileshe.net
[edit]
chileshe@chileshe-gw-01# commit
[edit]
chileshe@chileshe-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
chileshe@chileshe-gw-01#
```

Рис. 11: RA на eth1 (other-config-flag) и DHCPv6 common-options

Проверка конфигурации VyOS

A screenshot of a PuTTY terminal window titled "chileshe-gw-01 - PuTTY". The terminal displays a VyOS configuration script for a DHCPv6 server and a router advertisement. The configuration is as follows:

```
}  
}  
}  
dhcpv6-server {  
    shared-network-name chileshe-stateless {  
        common-options {  
            domain-search chileshe.net  
            name-server 2000::1  
        }  
        subnet 2000::0/64 {  
        }  
    }  
}  
router-advert {  
    interface eth1 {  
        other-config-flag  
        prefix 2000::/64 {  
        }  
    }  
}  
ssh {  
}  
}
```

The prompt is a green colon followed by a green cursor bar.

Проверка адресации на PC2 (SLAAC)

```
PC2-chileshe console is now available... Press RETURN to get started.
/ # ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
7: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN qlen 1000
    link/ether 02:42:4f:91:0c:00 brd ff:ff:ff:ff:ff:ff
    inet6 2000::42:4fff:fe91:c00/64 scope global dynamic flags 100
        valid_lft 2591971sec preferred_lft 14371sec
    inet6 fe80::42:4fff:fe91:c00/64 scope link
        valid_lft forever preferred_lft forever
/ # ip -6 route show
2000::/64 dev eth0 metric 256 expires 0sec
fe80::/64 dev eth0 metric 256
default via fe80::e56:73ff:fe6c:1 dev eth0 metric 1024 expires 0sec
/ # cat /etc/resolv.conf
/ #
/ # udhcpd6 -i eth0
udhcpd6: started, v1.37.0
udhcpd6: sending discover
udhcpd6: sending select
udhcpd6: no IAADDR option, ignoring packet
udhcpd6: sending select
udhcpd6: sending select
udhcpd6: sending select
udhcpd6: sending select
```

Рис. 13: IPv6-адрес из префикса 2000::/64 и маршрут по умолчанию

```
[edit]
chileshe@chileshe-gw-01#
[edit]
chileshe@chileshe-gw-01# run show dhcpv6 server leases
IPv6 address      State      Last communication      Lease expiration      Remaining
Type      Pool      IAID_DUID
-----
[edit]
chileshe@chileshe-gw-01#
```

Рис. 14: Получение DNS и domain-search (без назначения адреса)

Контроль на стороне сервера

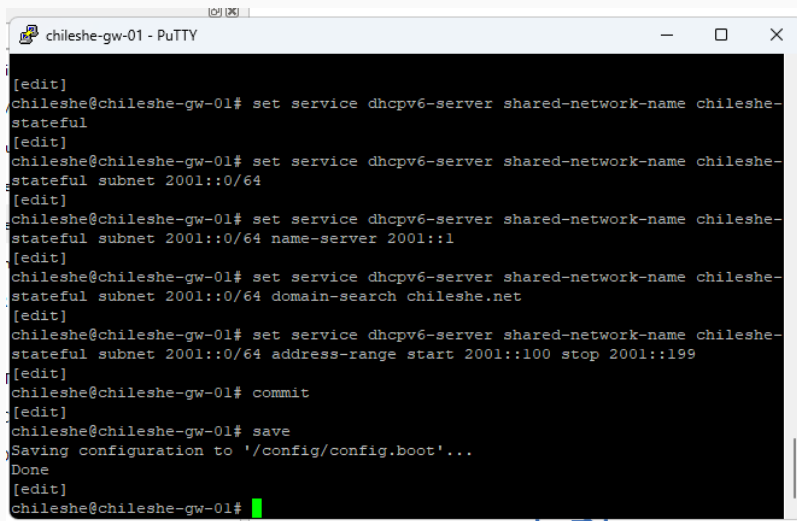
13	86.263234	fe80::42:4fff:fe91::ff02::1:2	ff02::1:2	DHCPv6	116 Solicit XID: 0xf4ad26 [ROOT-ONLY DOMAIN NAME][CID: 0003000102424f910c00
14	86.210263	fe80::e56:73ff:fe6c::fe80::42:4fff:fe91::	fe80::42:4fff:fe91::	DHCPv6	162 Advertise XID: 0xf4ad26 CID: 0003000102424f910c00
15	86.210419	fe80::42:4fff:fe91::fe80::e56:73ff:fe6c::	fe80::e56:73ff:fe6c::	ICMPv6	210 Destination Unreachable (Port unreachable)
16	86.227325	fe80::42:4fff:fe91::ff02::1:2	ff02::1:2	DHCPv6	134 Request XID: 0xf4ad26 [ROOT-ONLY DOMAIN NAME][CID: 0003000102424f910c00
17	86.227978	fe80::e56:73ff:fe6c::fe80::42:4fff:fe91::	fe80::42:4fff:fe91::	DHCPv6	162 Reply XID: 0xf4ad26 CID: 0003000102424f910c00
18	86.228082	fe80::42:4fff:fe91::fe80::e56:73ff:fe6c::	fe80::e56:73ff:fe6c::	ICMPv6	210 Destination Unreachable (Port unreachable)
19	89.243317	fe80::42:4fff:fe91::ff02::1:2	ff02::1:2	DHCPv6	182 Request XID: 0xf4ad26 [ROOT-ONLY DOMAIN NAME][CID: 0003000102424f910c00
20	89.244727	fe80::e56:73ff:fe6c::fe80::42:4fff:fe91::	fe80::42:4fff:fe91::	DHCPv6	162 Reply XID: 0xf4ad26 CID: 0003000102424f910c00
21	89.245207	fe80::42:4fff:fe91::fe80::e56:73ff:fe6c::	fe80::e56:73ff:fe6c::	ICMPv6	210 Destination Unreachable (Port unreachable)
22	91.459674	fe80::e56:73ff:fe6c::fe80::42:4fff:fe91::	fe80::42:4fff:fe91::	ICMPv6	86 Neighbor Solicitation for fe80::42:4fff:fe91:c00 from 0c:56:73:6c:00:01
23	91.460247	fe80::42:4fff:fe91::fe80::e56:73ff:fe6c::	fe80::e56:73ff:fe6c::	ICMPv6	78 Neighbor Advertisement fe80::42:4fff:fe91:c00 (sol)
24	91.572326	fe80::42:4fff:fe91::fe80::e56:73ff:fe6c::	fe80::e56:73ff:fe6c::	ICMPv6	86 Neighbor Solicitation for fe80::e56:73ff:fe6c:1 from 02:42:4f:91:0c:00
25	91.573680	fe80::e56:73ff:fe6c::fe80::42:4fff:fe91::	fe80::42:4fff:fe91::	ICMPv6	78 Neighbor Advertisement fe80::e56:73ff:fe6c:1 (rtr, sol)
26	92.260399	fe80::42:4fff:fe91::ff02::1:2	ff02::1:2	DHCPv6	182 Request XID: 0xf4ad26 [ROOT-ONLY DOMAIN NAME][CID: 0003000102424f910c00
27	92.261681	fe80::e56:73ff:fe6c::fe80::42:4fff:fe91::	fe80::42:4fff:fe91::	DHCPv6	162 Reply XID: 0xf4ad26 CID: 0003000102424f910c00
28	92.261935	fe80::42:4fff:fe91::fe80::e56:73ff:fe6c::	fe80::e56:73ff:fe6c::	ICMPv6	210 Destination Unreachable (Port unreachable)
29	95.273521	fe80::42:4fff:fe91::ff02::1:2	ff02::1:2	DHCPv6	182 Request XID: 0xf4ad26 [ROOT-ONLY DOMAIN NAME][CID: 0003000102424f910c00
30	95.278085	fe80::e56:73ff:fe6c::fe80::42:4fff:fe91::	fe80::42:4fff:fe91::	DHCPv6	162 Reply XID: 0xf4ad26 CID: 0003000102424f910c00

Message type: Solicit (1)	0000	33	33	00	01
Transaction ID: 0xf4ad26	0010	00	00	00	3e
Option: Elapsed time (8)	0020	4f	ff	fe	01
Length: 2	0030	00	00	00	01
Elapsed time: 0ms	0040	ad	26	00	08
Option Request	0050	00	00	00	00
Option: Identity Association for Non-temporary Address	0060	00	27	00	02
Option: Identity Association for Non-temporary Address (3)	0070	4f	91	0c	00
Length: 12					
IAID: 0691dc74					
T1: 0					
T2: 0					
Option Request					
Option: Option Request (6)					
Length: 4					
Requested Option code: DNS recursive name server (23)					
Requested Option code: Domain Search List (24)					
Client Fully Qualified Domain Name					
Option: Client Fully Qualified Domain Name (39)					
Length: 2					
Flags: 0x00 [CLIENT wants to update its AAAA RRs and SERVER to update its PTR RRs]					
Root only domain name: ['.'] (0)					
ERROR: A root-only domain name cannot be resolved.					
[Support Info (Server/Debian)]: ERROR: A root-only domain name cannot be resolved.					

Рис. 15: В Stateless аренды адресов отсутствуют

DHCPv6 Stateful

Настройка DHCPv6 Stateful на VyOS



```
[edit]
chileshe@chileshe-gw-01# set service dhcpv6-server shared-network-name chileshe-stateful
[edit]
chileshe@chileshe-gw-01# set service dhcpv6-server shared-network-name chileshe-stateful subnet 2001::0/64
[edit]
chileshe@chileshe-gw-01# set service dhcpv6-server shared-network-name chileshe-stateful subnet 2001::0/64 name-server 2001::1
[edit]
chileshe@chileshe-gw-01# set service dhcpv6-server shared-network-name chileshe-stateful subnet 2001::0/64 domain-search chileshe.net
[edit]
chileshe@chileshe-gw-01# set service dhcpv6-server shared-network-name chileshe-stateful subnet 2001::0/64 address-range start 2001::100 stop 2001::199
[edit]
chileshe@chileshe-gw-01# commit
[edit]
chileshe@chileshe-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
chileshe@chileshe-gw-01#
```

Рис. 16: Пул адресов 2001::100–2001::199, DNS 2001::1, domain-search

Проверка PC3 до выдачи адреса

```
PC3-chileshe console is now available... Press RETURN to get started.  
/ # ip address  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
8: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN  
    qlen 1000  
    link/ether 02:42:0f:7e:94:00 brd ff:ff:ff:ff:ff:ff  
    inet6 fe80::42:fff:fe7e:9400/64 scope link  
        valid_lft forever preferred_lft forever  
/ # ip -6 route show  
fe80::/64 dev eth0 metric 256  
default via fe80::e56:73ff:fe6c:2 dev eth0 metric 1024 expires 0sec  
/ # cat /etc/resolv.conf  
/ # udhcpc6 -i eth0  
udhcpc6: started, v1.37.0  
udhcpc6: sending discover  
udhcpc6: sending select  
udhcpc6: IPv6 obtained, lease time 43200  
/ #
```

Рис. 17: Начальные параметры сети и маршруты IPv6

Получение IPv6-адреса по DHCPv6

```
/ #
/ # ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
8: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN qlen 1000
    link/ether 02:42:0f:7e:94:00 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::42:fff:fe7e:9400/64 scope link
        valid_lft forever preferred_lft forever
/ # ip -6 route show
fe80::/64 dev eth0 metric 256
default via fe80::e56:73ff:fe6c:2 dev eth0 metric 1024 expires 0sec
/ # ping 2001::1 -c 2
PING 2001::1 (2001::1): 56 data bytes
64 bytes from 2001::1: seq=0 ttl=64 time=0.890 ms
64 bytes from 2001::1: seq=1 ttl=64 time=0.845 ms

--- 2001::1 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.845/0.867/0.890 ms
/ # cat /etc/resolv.conf
search chileshe.net
nameserver 2001:0000:0000:0000:0000:0000:0000:0001
/ #
```

Рис. 18: dhclient -6 -v: адрес получен, зафиксировано время аренды


```
[edit]
chileshe@chileshe-gw-01# run show dhcpv6 server leases
IPv6 address      State      Last communication      Lease expiration      Remaining
Type      Pool      IAID_DUID
-----
[edit]
chileshe@chileshe-gw-01#
[edit]
chileshe@chileshe-gw-01# run show dhcpv6 server leases
IPv6 address      State      Last communication      Lease expiration      Remaining
Type      Pool      IAID_DUID
-----
2001::199      active      2025/12/16 08:59:33      2025/12/16 20:59:33      11:58:44
non-temporary      chileshe-stateful      25:37:34:f5:00:03:00:01:02:42:0f:7e:94:00
[edit]
chileshe@chileshe-gw-01#
```

Рис. 19: ping 2001::1 и параметры resolv.conf

Выданные аренды DHCPv6 на маршрутизаторе

No.	Time	Source	Destination	Protocol	Length	Info
8	4.092078	fe80::e56:73ff:fe6c...	ff02::1	ICMPv6	86	Router Advertisement from 0c:56:73:6c:00:02
9	7.140424	fe80::e56:73ff:fe6c...	fe80::42:fff:fe7e:9...	ICMPv6	86	Neighbor Solicitation for fe80::42:fff:fe7e:9400 from 0c:56:73:6c:00:02
10	7.140613	fe80::42:fff:fe7e:9...	fe80::e56:73ff:fe6c...	ICMPv6	78	Neighbor Advertisement fe80::42:fff:fe7e:9400 (sol)
11	12.399578	fe80::42:fff:fe7e:9...	fe80::e56:73ff:fe6c...	ICMPv6	86	Neighbor Solicitation for fe80::e56:73ff:fe6c:2 from 02:42:0f:7e:94:00
12	12.401238	fe80::e56:73ff:fe6c...	fe80::42:fff:fe7e:9...	ICMPv6	78	Neighbor Advertisement fe80::e56:73ff:fe6c:2 (rtr, sol)
13	59.852854	fe80::42:fff:fe7e:9...	ff02::1:2	DHCPv6	116	Solicit XID: 0xb28333 [ROOT-ONLY DOMAIN NAME]CID: 0003000102420f7e9400
14	59.863266	fe80::e56:73ff:fe6c...	fe80::42:fff:fe7e:9...	DHCPv6	180	Advertise XID: 0xb28333 IAA: 2001:199 CID: 0003000102420f7e9400
15	59.863364	fe80::42:fff:fe7e:9...	fe80::e56:73ff:fe6c...	ICMPv6	228	Destination Unreachable (Port unreachable)
16	59.885838	fe80::42:fff:fe7e:9...	ff02::1:2	DHCPv6	134	Request XID: 0xb28333 [ROOT-ONLY DOMAIN NAME]CID: 0003000102420f7e9400
17	59.886730	fe80::e56:73ff:fe6c...	fe80::42:fff:fe7e:9...	DHCPv6	180	Reply XID: 0xb28333 IAA: 2001:199 CID: 0003000102420f7e9400
18	59.886968	fe80::42:fff:fe7e:9...	fe80::e56:73ff:fe6c...	ICMPv6	228	Destination Unreachable (Port unreachable)
19	64.999059	fe80::e56:73ff:fe6c...	fe80::42:fff:fe7e:9...	ICMPv6	86	Neighbor Solicitation for fe80::42:fff:fe7e:9400 from 0c:56:73:6c:00:02
20	64.999702	fe80::42:fff:fe7e:9...	fe80::e56:73ff:fe6c...	ICMPv6	78	Neighbor Advertisement fe80::42:fff:fe7e:9400 (sol)
21	65.136503	fe80::42:fff:fe7e:9...	fe80::e56:73ff:fe6c...	ICMPv6	86	Neighbor Solicitation for fe80::e56:73ff:fe6c:2 from 02:42:0f:7e:94:00
22	65.138803	fe80::e56:73ff:fe6c...	fe80::42:fff:fe7e:9...	ICMPv6	78	Neighbor Advertisement fe80::e56:73ff:fe6c:2 (rtr, sol)
23	104.397537	fe80::42:fff:fe7e:9...	2001::1	ICMPv6	118	Echo (ping) request id=0x013a, seq=0, hop limit=64 (reply in 24)
24	104.398309	2001::1	fe80::42:fff:fe7e:9...	ICMPv6	118	Echo (ping) reply id=0x013a, seq=0, hop limit=64 (request in 23)
25	105.398053	fe80::42:fff:fe7e:9...	2001::1	ICMPv6	118	Echo (ping) request id=0x013a, seq=1, hop limit=64 (reply in 26)
26	105.398692	2001::1	fe80::42:fff:fe7e:9...	ICMPv6	118	Echo (ping) reply id=0x013a, seq=1, hop limit=64 (request in 25)
27	109.761067	fe80::e56:73ff:fe6c...	fe80::42:fff:fe7e:9...	ICMPv6	86	Neighbor Solicitation for fe80::42:fff:fe7e:9400 from 0c:56:73:6c:00:02
28	109.761178	fe80::42:fff:fe7e:9...	fe80::e56:73ff:fe6c...	ICMPv6	78	Neighbor Advertisement fe80::42:fff:fe7e:9400 (sol)
29	110.423025	fe80::42:fff:fe7e:9...	fe80::e56:73ff:fe6c...	ICMPv6	86	Neighbor Solicitation for fe80::e56:73ff:fe6c:2 from 02:42:0f:7e:94:00
30	110.437676	fe80::e56:73ff:fe6c...	fe80::42:fff:fe7e:9...	ICMPv6	78	Neighbor Advertisement fe80::e56:73ff:fe6c:2 (rtr, sol)

User Datagram Protocol, Src Port: 547, Dst Port: 546
Source Port: 547
Destination Port: 546
Length: 126
Checksum: 0x06f2 [unverified]
[Checksum Status: Unverified]
[Stream Index: 1]
[Stream Packet Number: 2]
[Timestamps]
UDP payload (118 bytes)

DHCPv6
Message type: Reply (7)
Transaction ID: 0xb28333
Identity Association for Non-temporary Address
Client Identifier
Option: Client Identifier (1)
Length: 10
DUID: 0003000102420f7e9400
DUID Type: link-layer address (3)
Hardware type: Ethernet (1)
Link-layer address: 02:42:0f:7e:94:00
Link-layer address (Ethernet): 02:42:0f:7e:94:00 (02:42:0f:7e:94:00)
Server Identifier
Option: Server Identifier (2)
Length: 14
DUID: 0001000130d3d90e0c56736c0001
DUID Type: link-layer address plus time (1)
Hardware type: Ethernet (1)
DUID Time: Dec 16, 2025 11:47:10.000000000 RTZ 2 (sumo)
Link-layer address: 0c:56:73:6c:00:01

0000 02 42 0f 7e
0010 9b 87 00 7e
0020 73 ff fe 6c
0030 0f ff fe 7e
0040 83 33 00 03
0050 00 00 00 05
0060 00 00 00 00
0070 00 0a 00 03
0080 00 01 00 01
0090 00 10 20 01
00a0 00 01 00 18
00b0 6e 65 74 00

Вывод

- Выполнена настройка DHCP для IPv4 на VyOS, подтверждена автоматическая выдача адреса и параметров сети клиенту.
- Реализованы два режима DHCPv6:
 - Stateless: IPv6-адрес формируется SLAAC, DHCPv6 передаёт дополнительные параметры (DNS, domain-search).
 - Stateful: IPv6-адрес назначается DHCPv6-сервером из заданного пула.
- Анализ захваченного трафика подтвердил корректную последовательность сообщений DHCP/DHCPv6 и работы Router Advertisements.